

# Introduction

*Benoît Legat*

*Understand simple things deeply*

☐ Full Width Mode    ☐ Present Mode

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# Deepseek event ⇄

## Nvidia stocks recovered slightly on Tuesday

Nvidia share price



Note: Axis does not start at zero. Markets are open Mon-Fri 9:30-16:00 Eastern time

Source: Bloomberg. Last update: 28 Jan 2025

B B C

# Distillation ⇄

## 2.3.1. Cold Start

Unlike DeepSeek-R1-Zero, to prevent the early unstable cold start phase of RL training from the base model, for DeepSeek-R1 we construct and collect a small amount of long CoT data to fine-tune the model as the initial RL actor. To collect such data, we have explored several approaches: using few-shot prompting with a long CoT as an example, directly prompting models to generate detailed answers with reflection and verification, gathering DeepSeek-R1-Zero outputs in a readable format, and refining the results through post-processing by human annotators.

In this work, we collect thousands of cold-start data to fine-tune the DeepSeek-V3-Base as the starting point for RL. Compared to DeepSeek-R1-Zero, the advantages of cold start data

| Model                        | AIME 2024 |         | MATH-500 | GPQA Diamond | LiveCodeBench |
|------------------------------|-----------|---------|----------|--------------|---------------|
|                              | pass@1    | cons@64 | pass@1   | pass@1       | pass@1        |
| QwQ-32B-Preview              | 50.0      | 60.0    | 90.6     | 54.5         | 41.9          |
| DeepSeek-R1-Zero-Qwen-32B    | 47.0      | 60.0    | 91.6     | 55.0         | 40.2          |
| DeepSeek-R1-Distill-Qwen-32B | 72.6      | 83.3    | 94.3     | 62.1         | 57.2          |

Table 6 | Comparison of distilled and RL Models on Reasoning-Related Benchmarks.

# Low-level improvements ⇄

.....

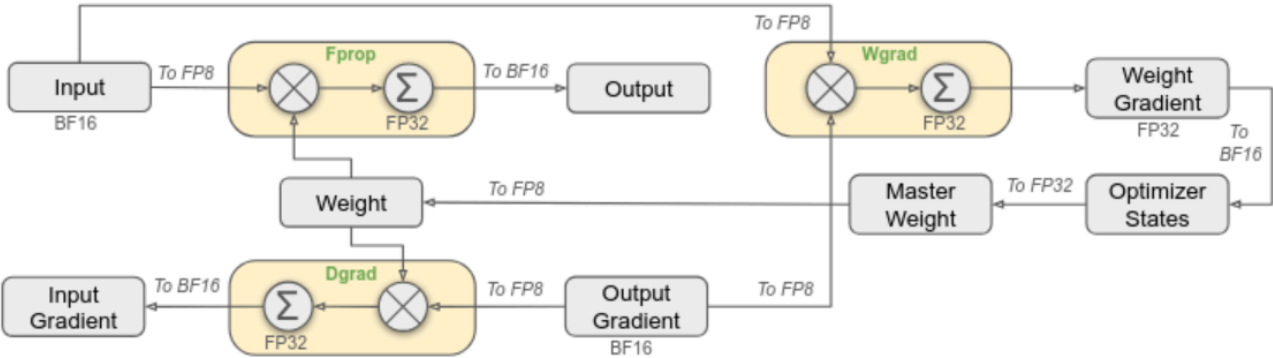


Figure 6 | The overall mixed precision framework with FP8 data format. For clarification, only the Linear operator is illustrated.

► Is it worth developing such low-level improvements?

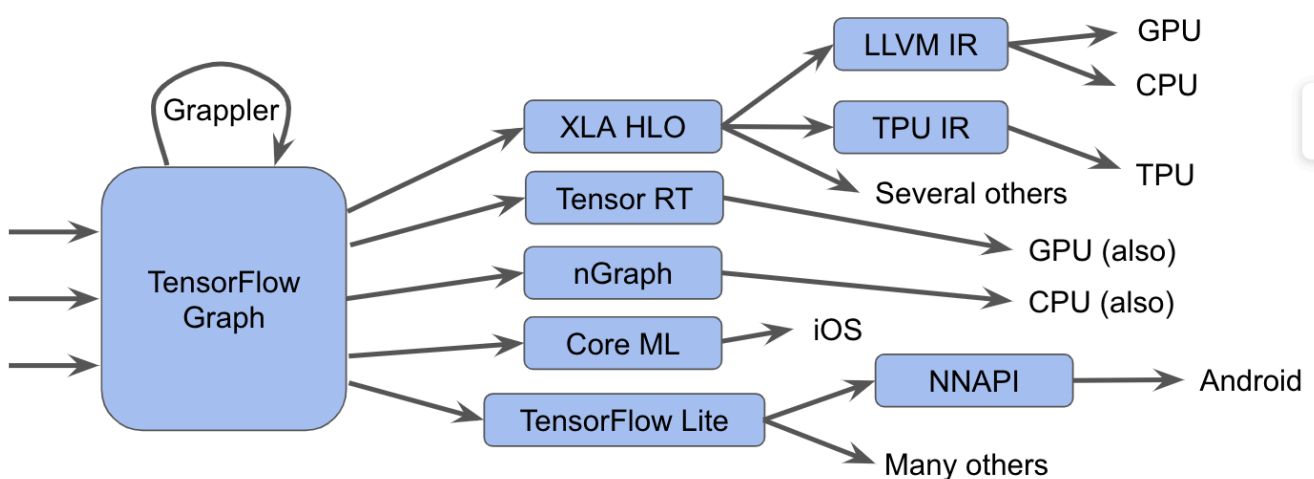
# What do these libraries really do ? ⇌

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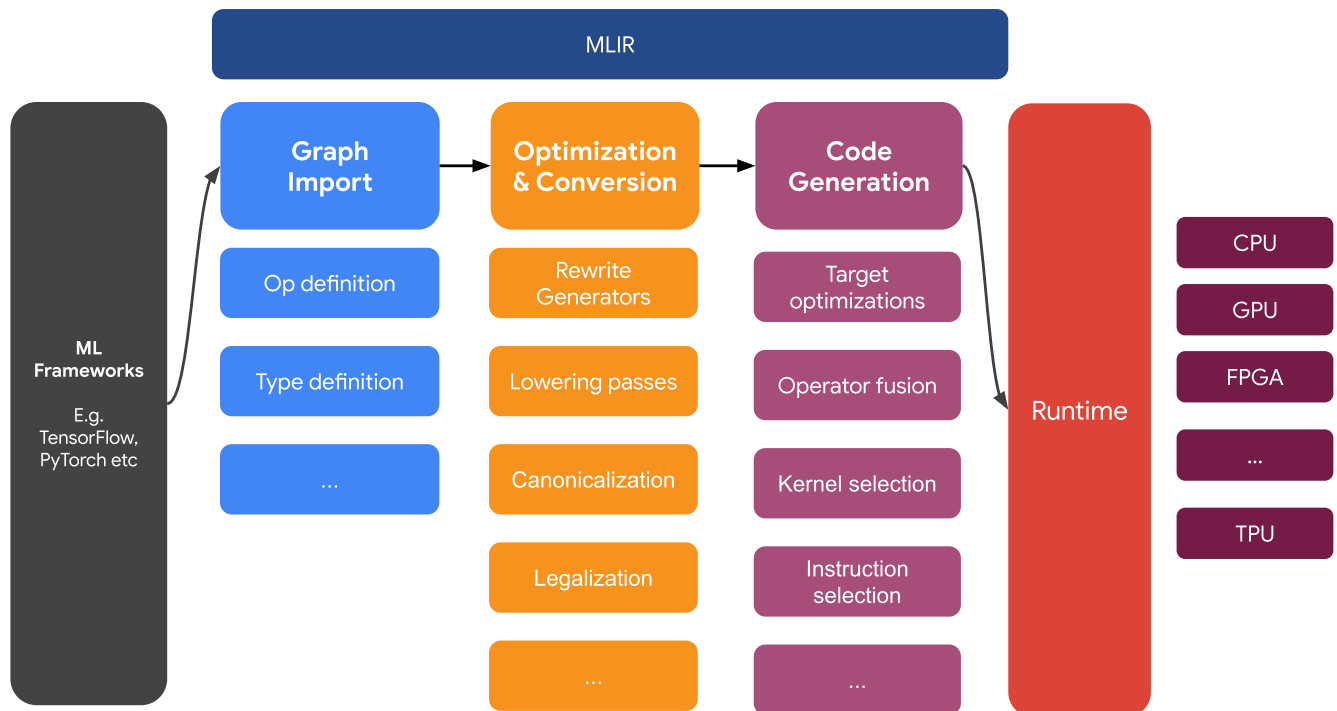
## Accelerated automatic differentiation ⇌

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## MLIR ↻

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## DIY ↻

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The goal of the course is to take a tour of **automatic differentiation**. For this we will write our own from scratch, in **julia**.

Yann LeCun  

@ylecun · [Follow](#)



Hotter take: ML would have advanced faster if another front-end language had been available and widely adopted instead of Python.

One that is interactive yet fast & compilable, multithreaded (no GIL), isn't bloated, doesn't care about white spaces,... E.g. Julia or some Lisp.

**Bojan Tunguz**  @tunguz

Hot take: Machine Learning would not have been nearly as advanced now were it not for Python. Python's two main virtues in the context of ML:

1. Lowering barriers to entry.
2. As a scripting language, it encourages and enables experimental workflow.

1:27 PM · Feb 22, 2023



3.2K



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# Grading ⇄

- If exam or homework is below 5, grade is the minimum of both
- If exam and homework are above 10, grade is the average of both
- Otherwise, we interpolate between these cases as follows
- Gain bonus points by contributing to the Git or winning benchmarks with the projects

f (generic function with 1 method)

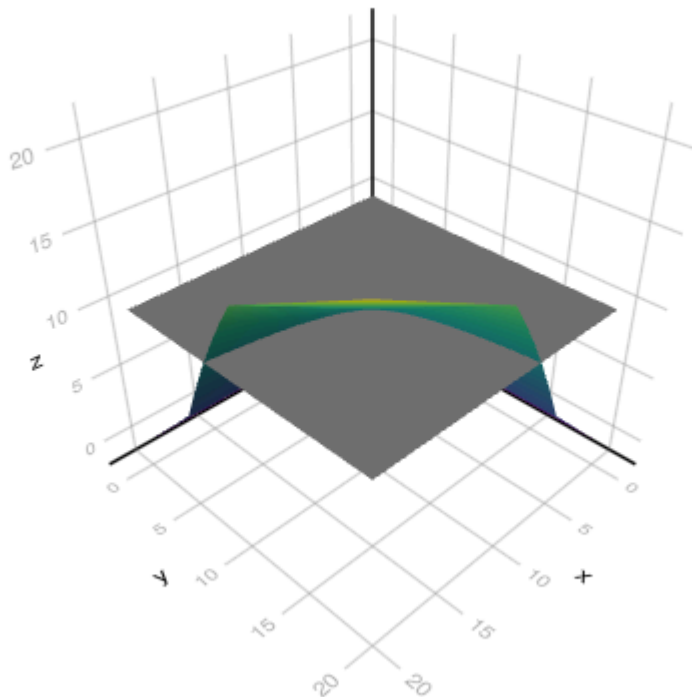
```
1 f(x) = min(max(0, (x - 5)/10), 0.5)
```

g (generic function with 1 method)

```
1 g(a, b) = f(a) * b + (1 - f(a)) * a
```

grade (generic function with 1 method)

```
1 grade(HW, EX) = min(g(HW, EX), g(EX, HW))
```



The End

```
1 using Colors, WGLMakie
```

```
1 using Plots, PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral; @html, @html_str  
   PlutoTeachingTools, ShortCodes
```

img (generic function with 3 methods)

qa (generic function with 2 methods)